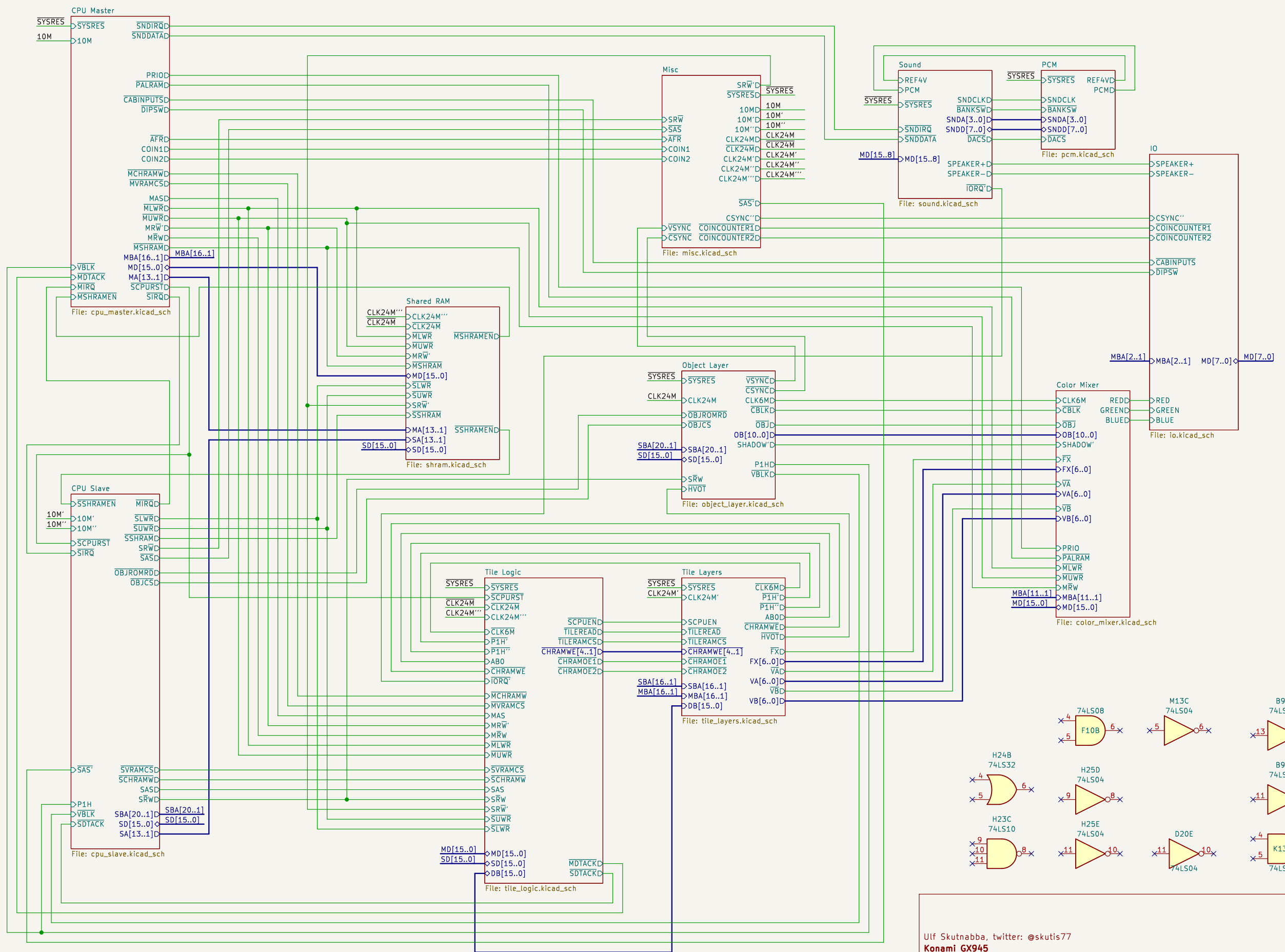




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Sheet: /
File: grad3.kicad_sch

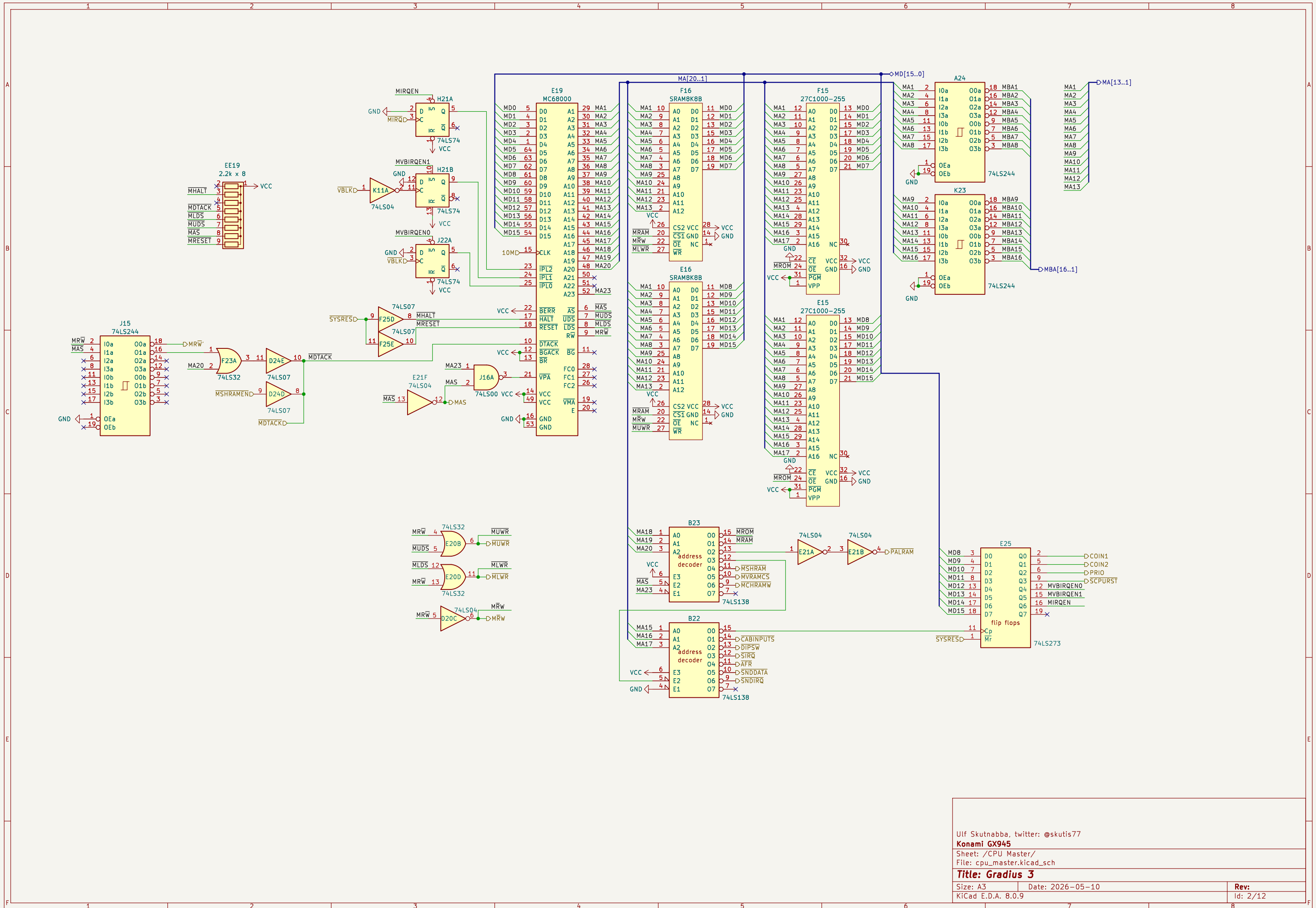
Title: Gradius 3

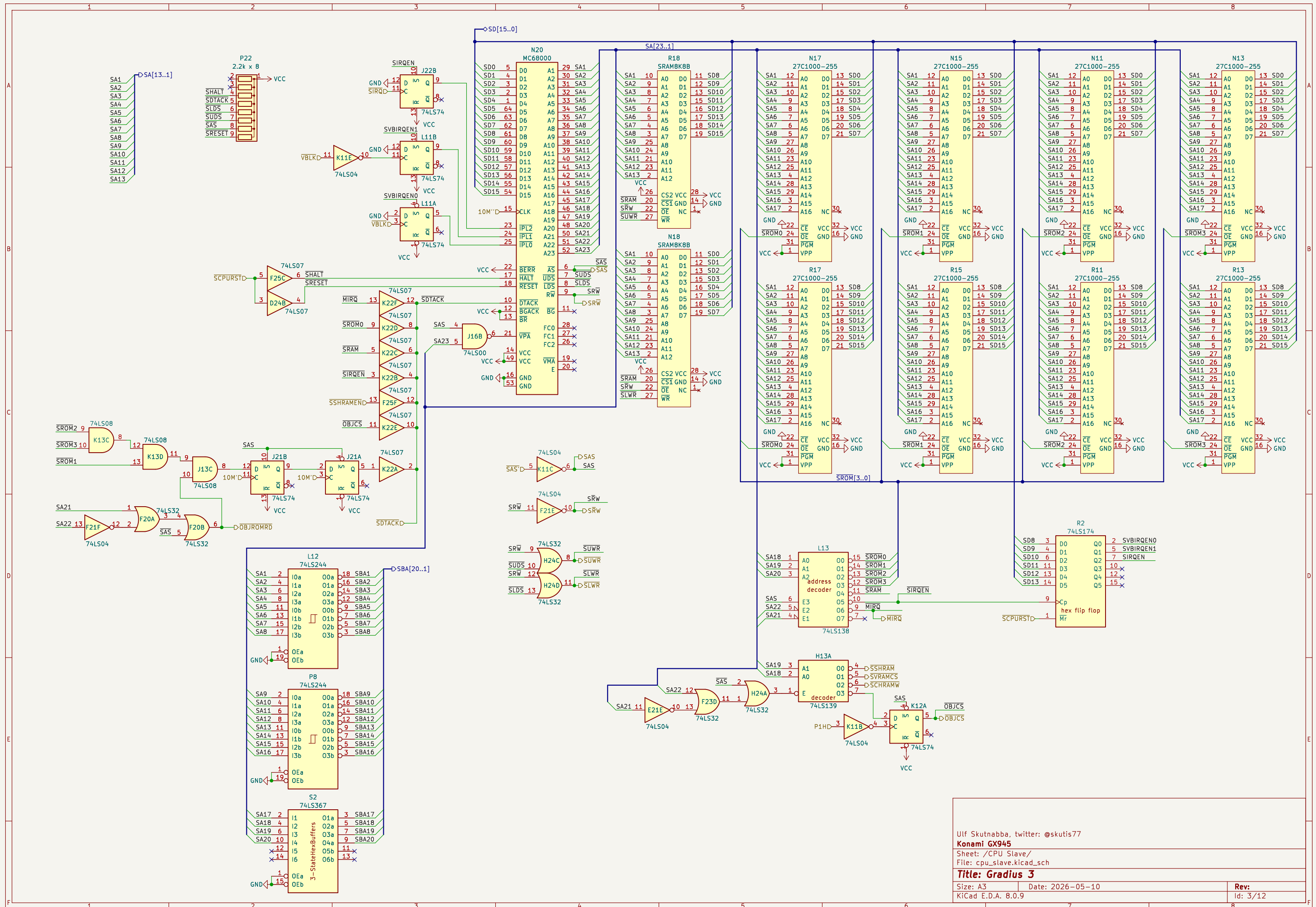
Size: A3	Date: 2026-05-10
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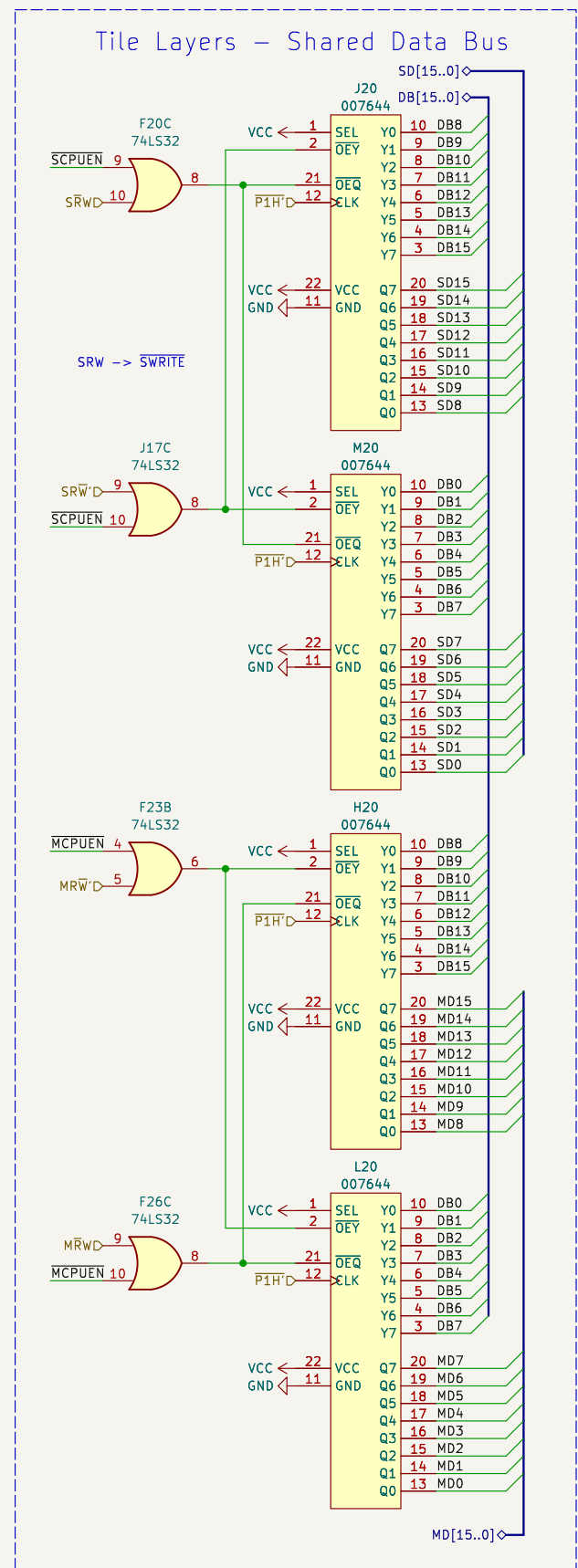
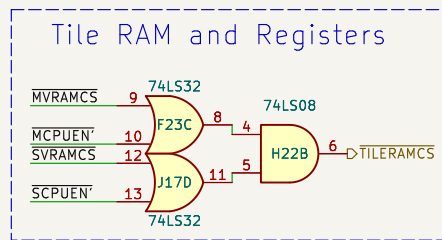
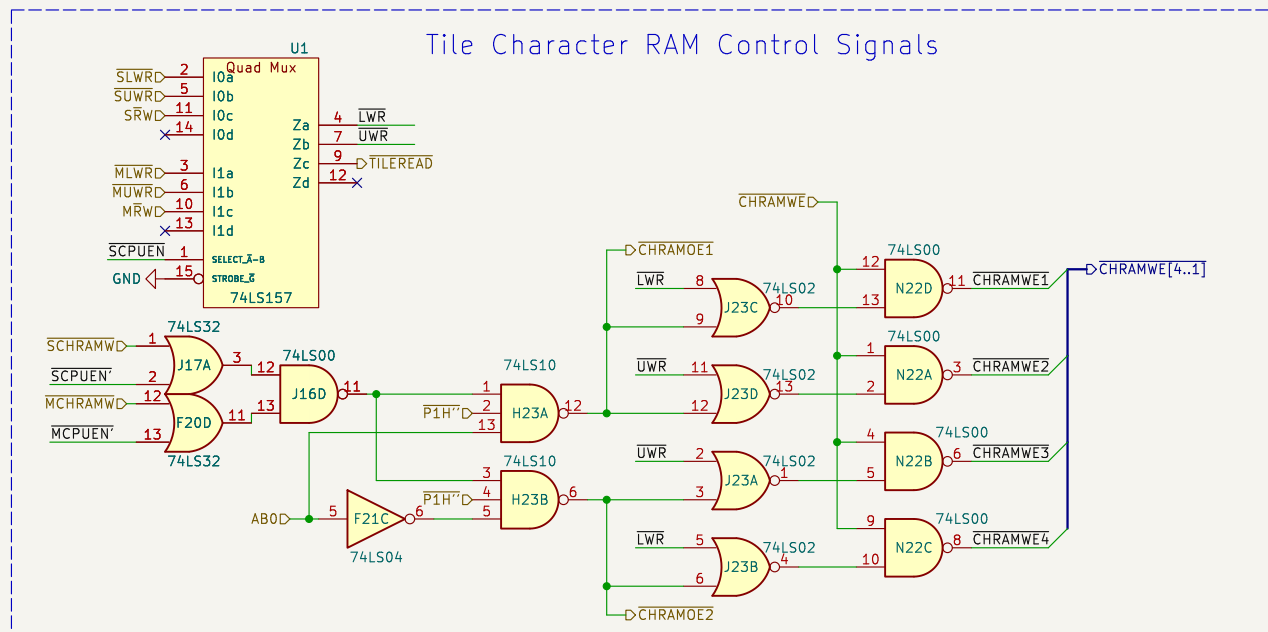
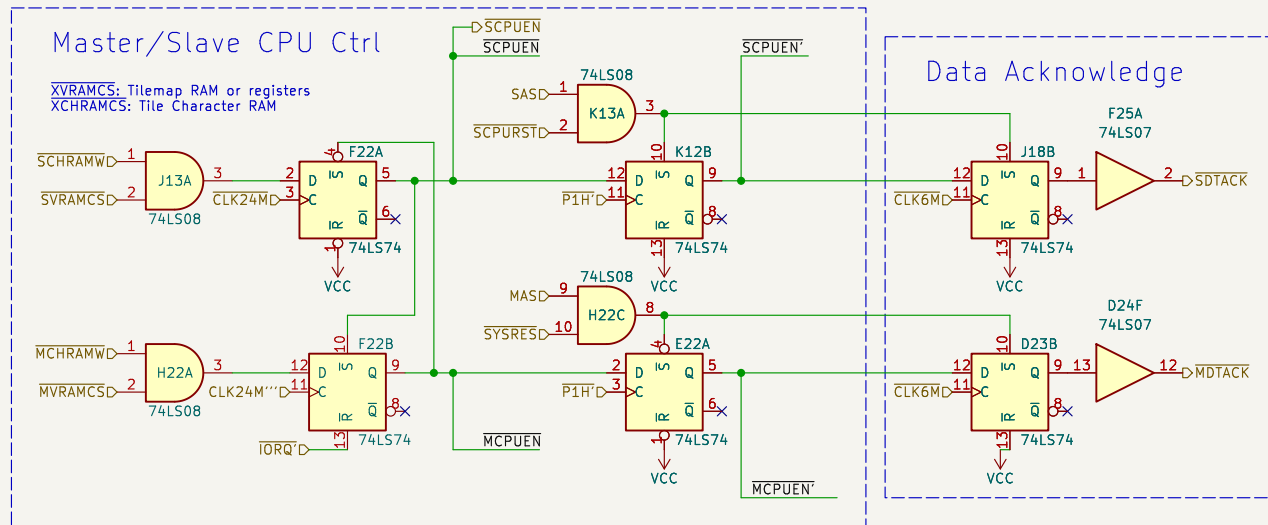
	7
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	7	8





7		8
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Sheet: /Tile Logic/

File: tile_logic.kicad_sch

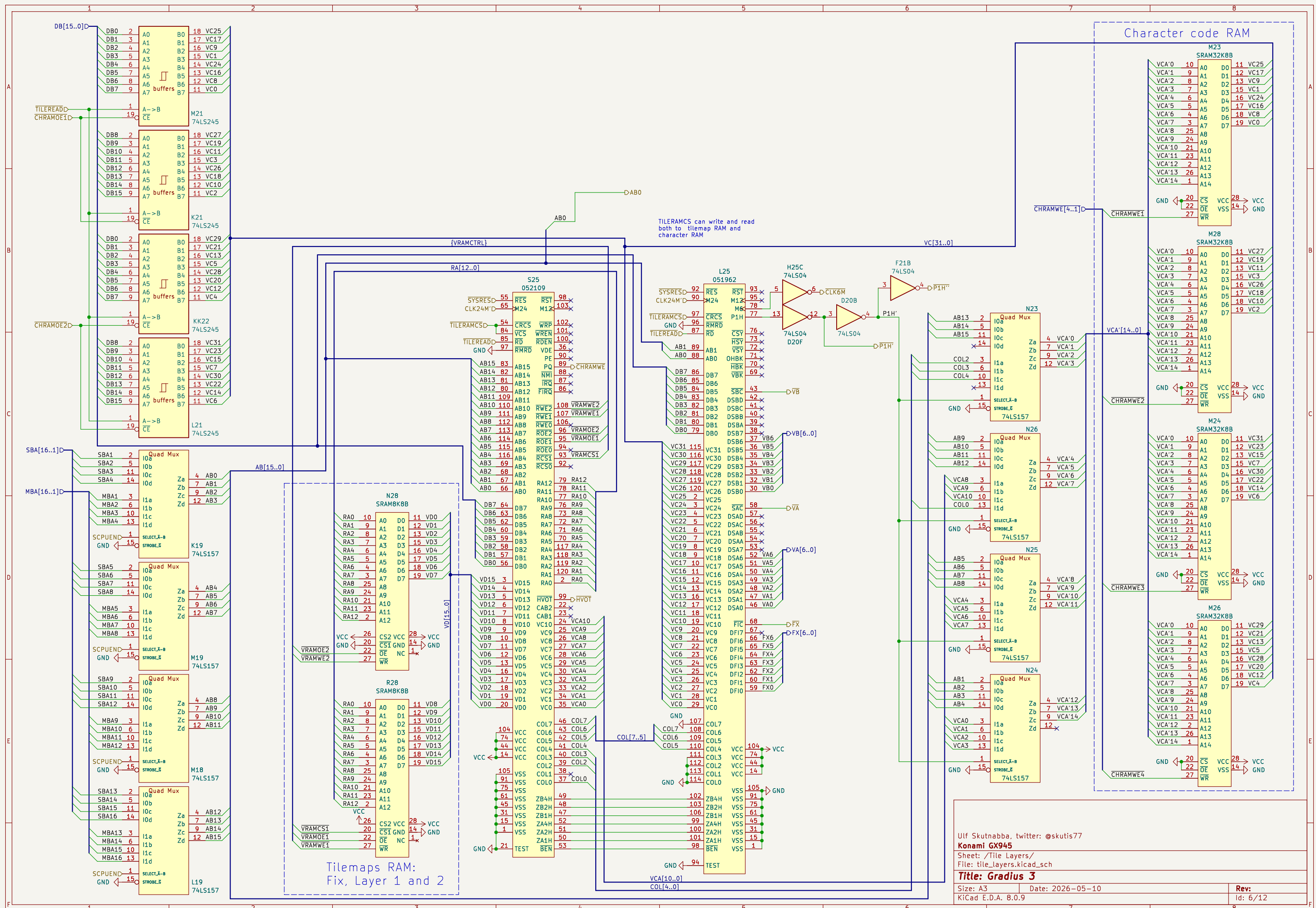
Title: Gradius 3

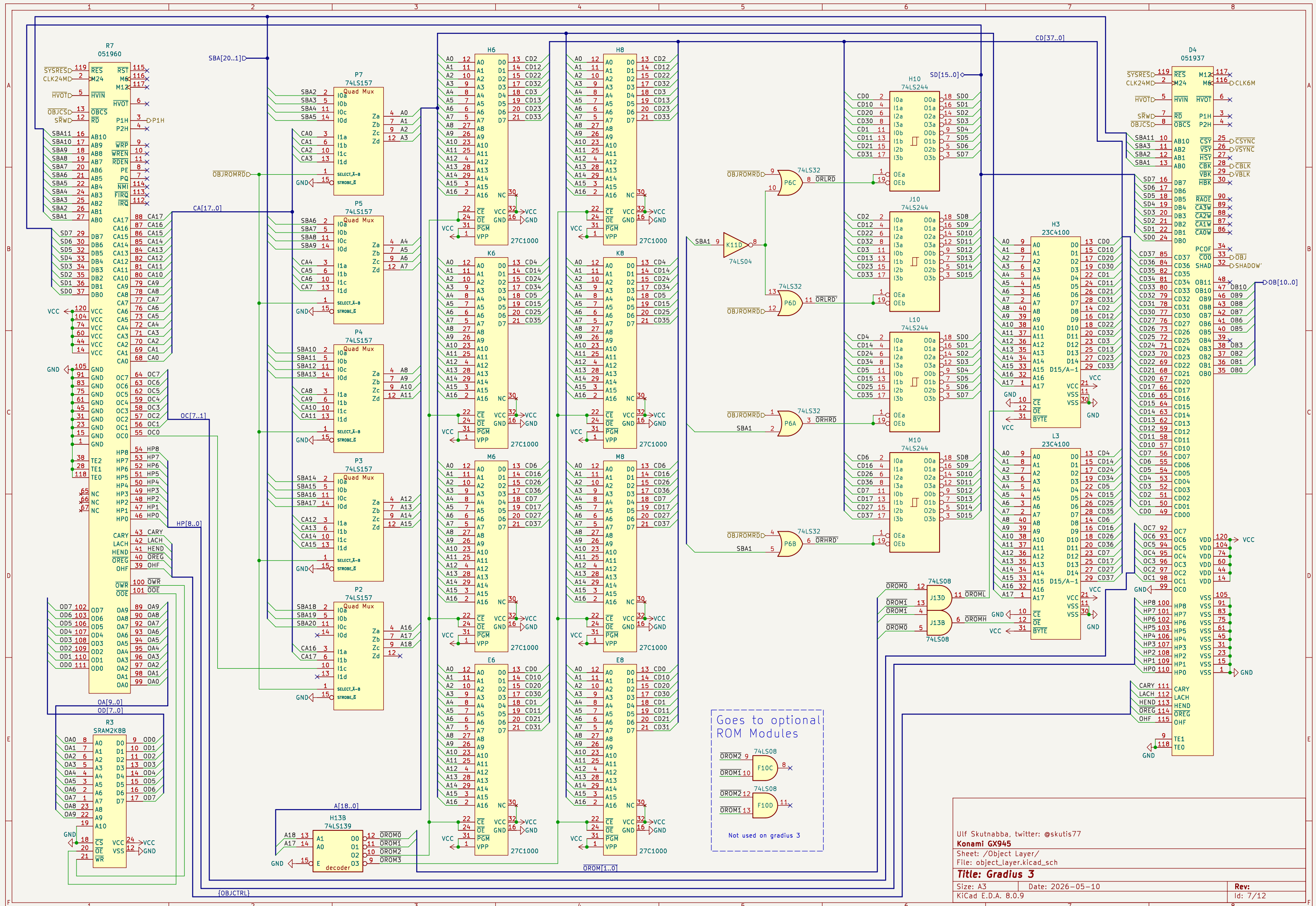
Size: A3 Date: 2026-05-10

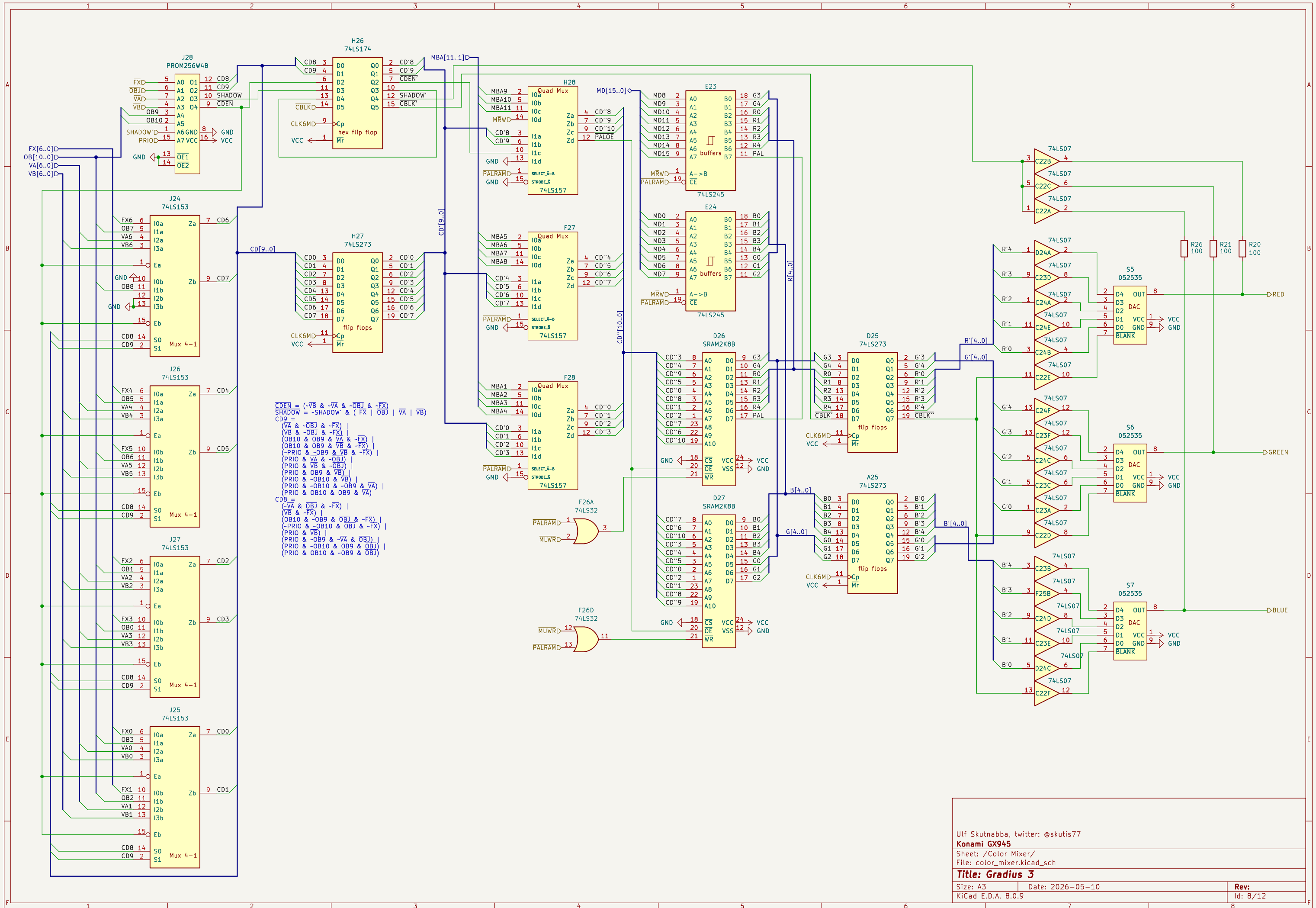
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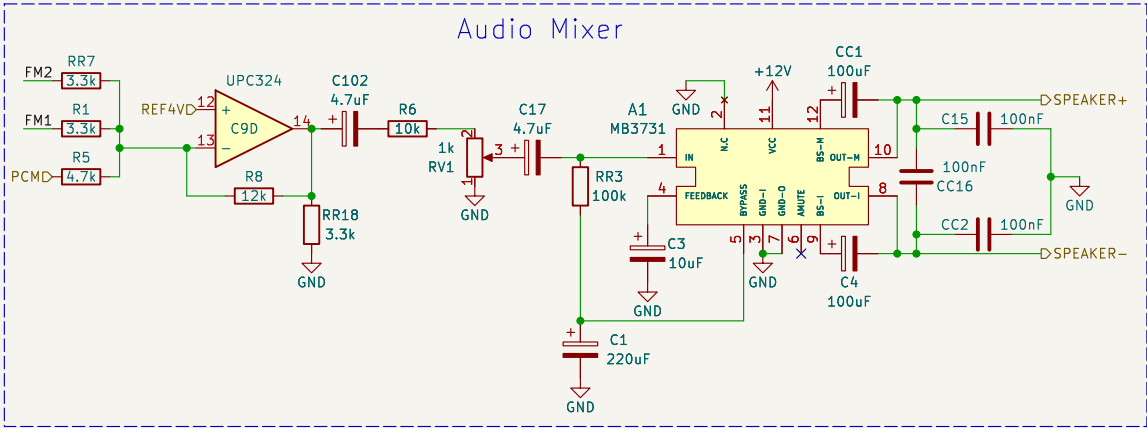
Rev:

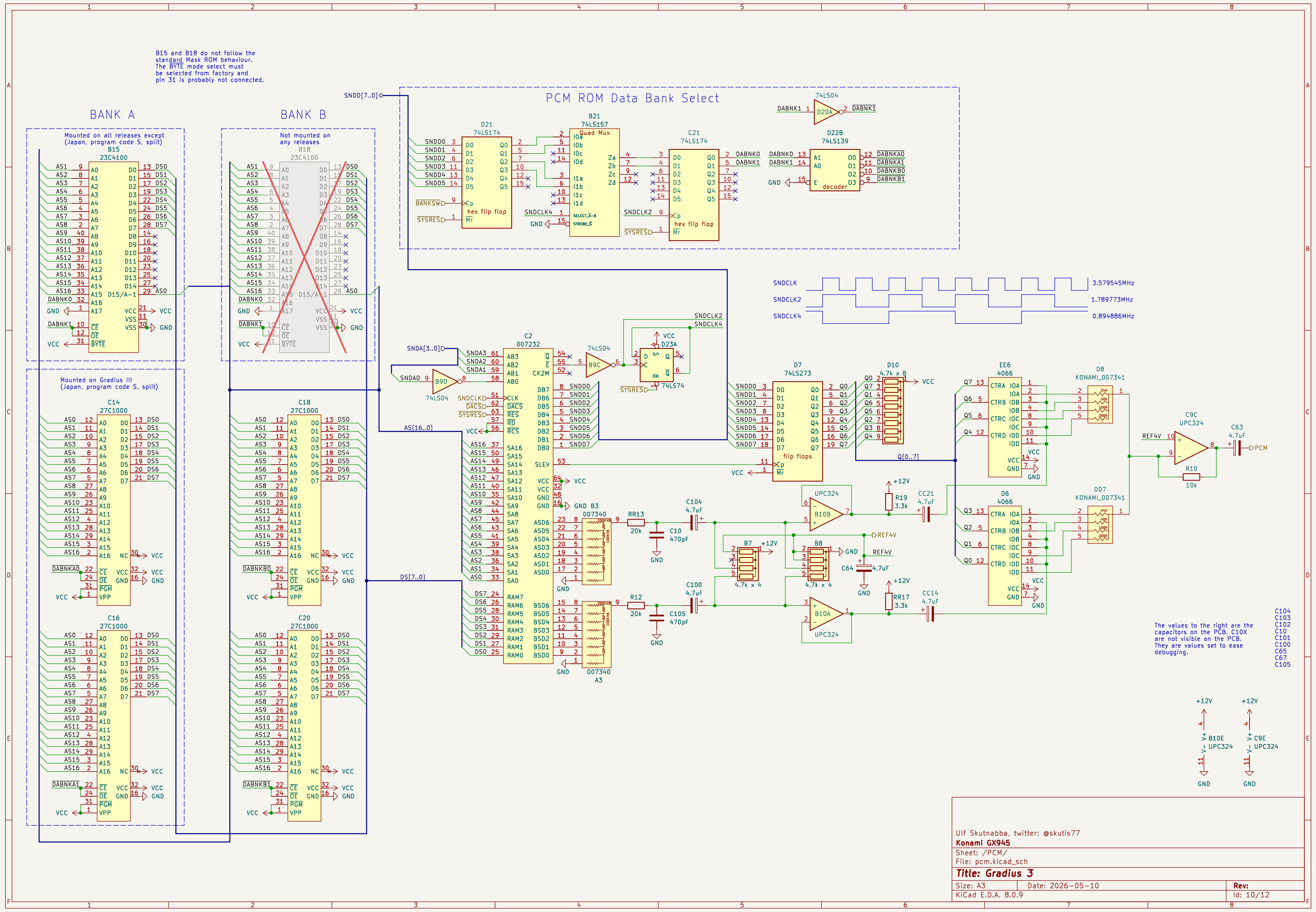
Id: 5/12



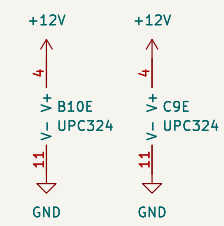








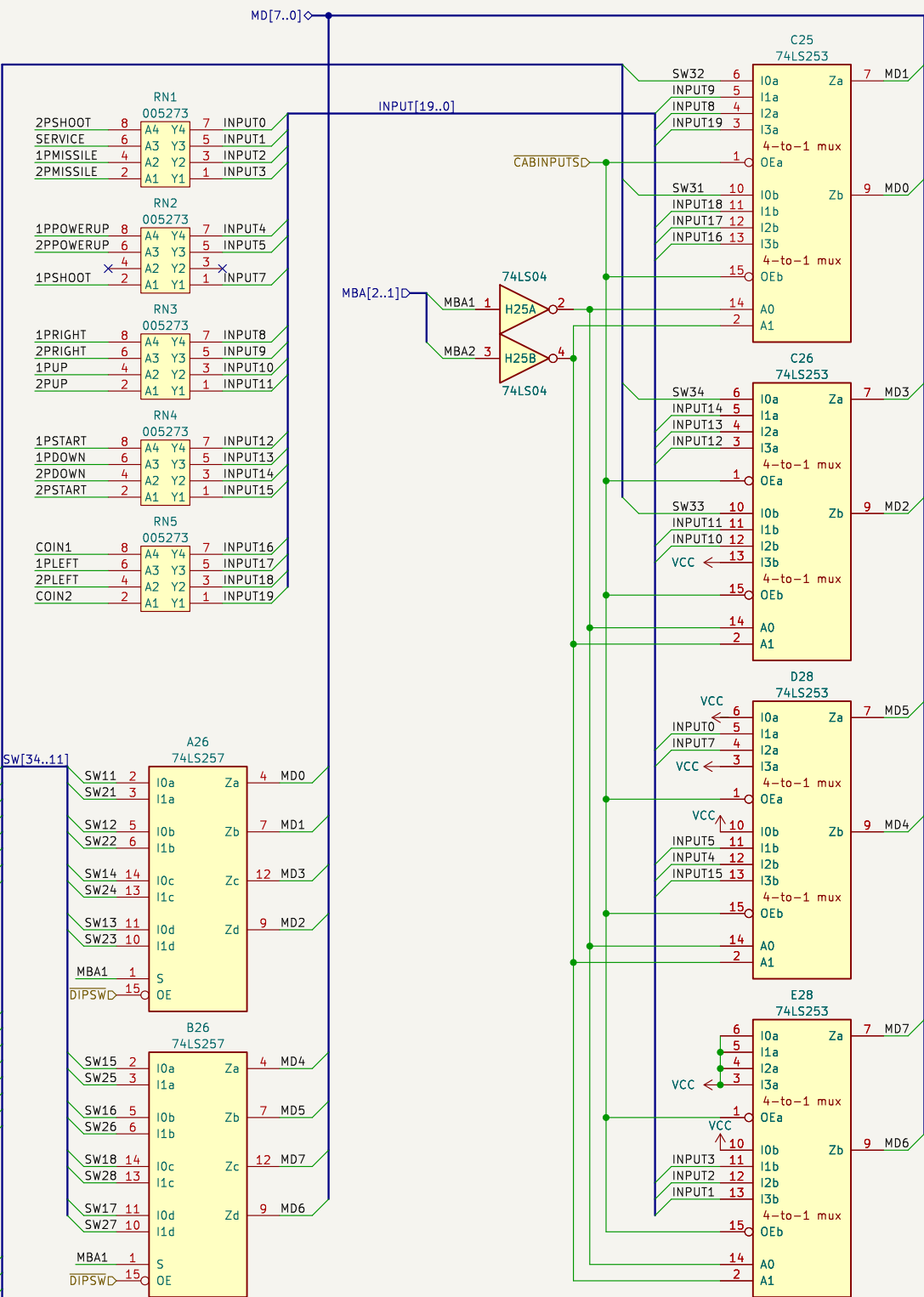
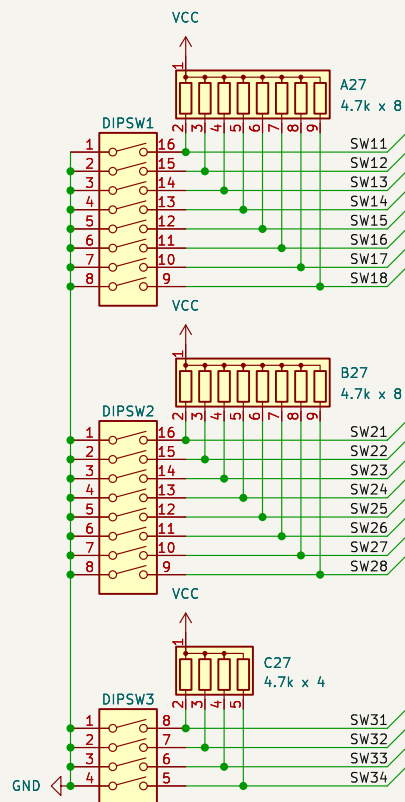
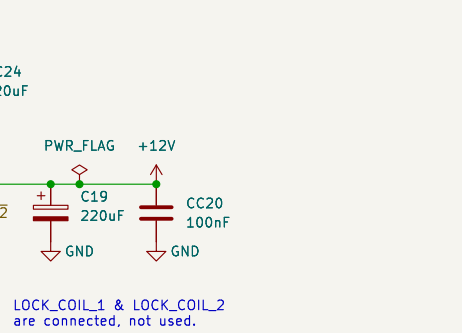
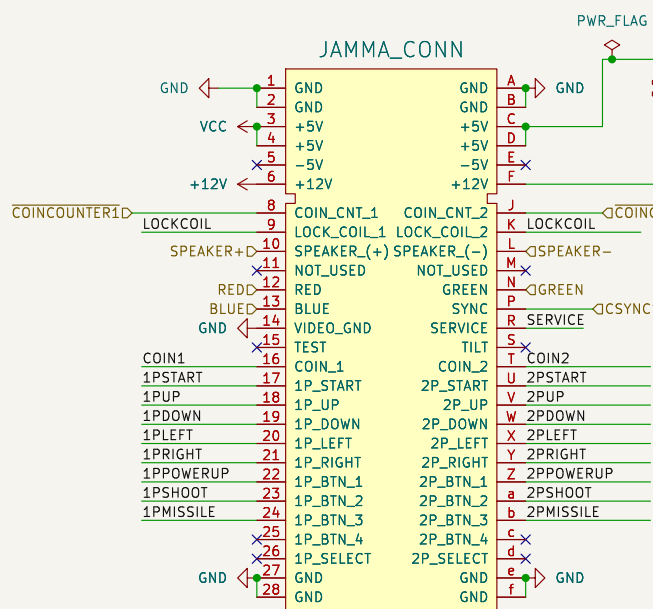
C104
C103
C102
C10
C101
C100
C65
C67
C105



Size: A3	Da
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	7	8
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In the schematic symbol for LS257 the channels are in wrong order but the connections are correct.

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Sheet: /10/

File: io.kicad_sch

Title: Gradius 3

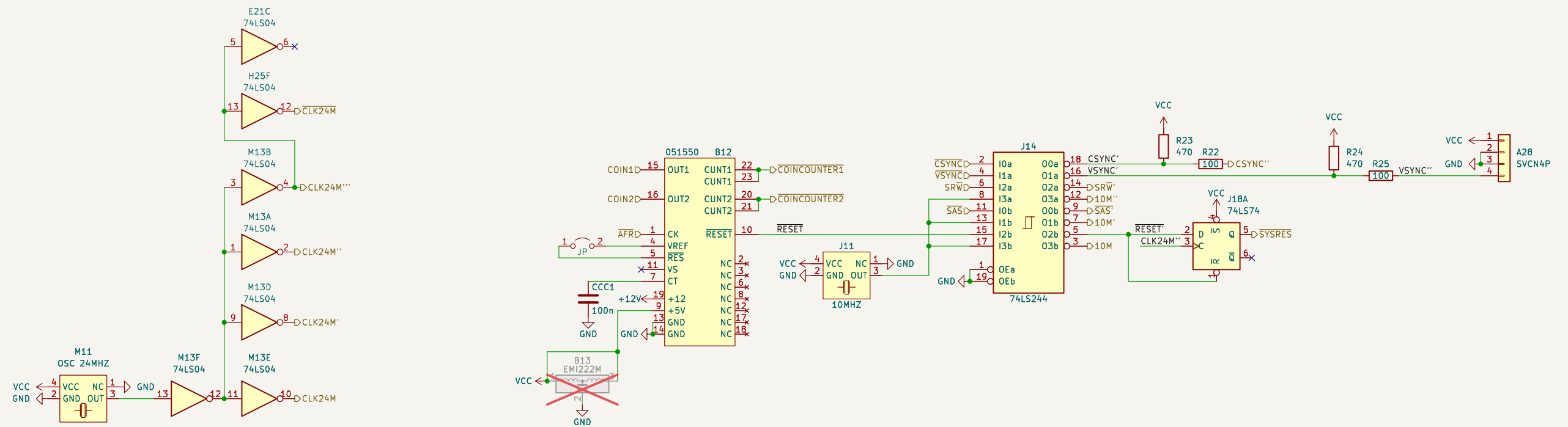
Size: A3

Date: 2026-05-10

Rev:

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Id: 11/12



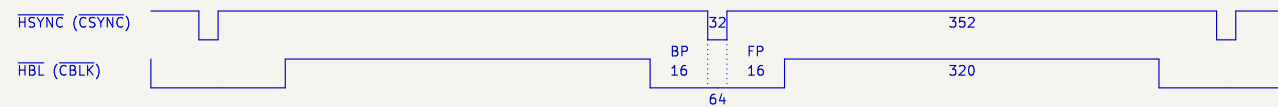
Horizontal and vertical synch timing diagrams

The pixel clock is derived from the 24MHz oscillator.
Pixel clock OVCK: $f = 24\text{MHz} / 4 = 6\text{MHz}$

The numbers in the HSYNC and HBL diagram are pixel clock cycles.
All edges are synchronised to the rising edge of the pixel clock.

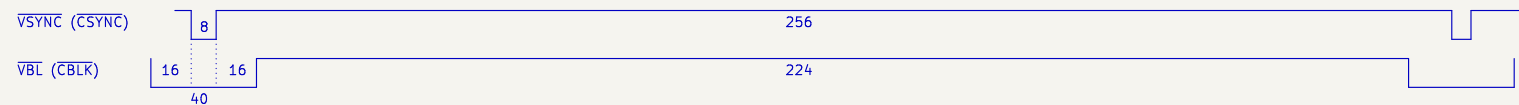
The signals have been measured at the output of the graphic chips.

If horizontal blanking is measured at the RGB DACs, the blanking is delayed 2 pixel clocks relative to composite sync. This gives BP = 14 and FP = 18.



HSYNC and HBL
Frequency $f = 6\text{MHz} / 384 = 15.625\text{kHz}$.
Period $T = 1/f = 64\mu\text{s}$.

The numbers in the VSYNC and VBL diagram are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



VSYNC and VBL:
Frequency $f = 15.625\text{kHz} / 264 = 59.1856\text{Hz}$
Period $T = 1/f = 1 / 59.1856\text{Hz} = 16.896\text{ms}$

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Sheet: /Misc/

File: misc.kicad_sch

Title: Gradius 3

Size: A3

Date: 2026-05-10

Rev:

KiCad E.D.A. 8.0.9

Id: 12/12